RAJALAKSHMI ENGINEERING COLLEGE

Canci Bricilla A 251001036-2025-IT

C2

Started on	Thursday, 20 August 2026, 10:36 AM
State	Finished
Completed on	Thursday, 20 August 2026, 10:37 AM
Time taken	1 min 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the floor of x using binary search (Divide & Conquer)
int findFloor(int arr[], int n, int x) {
    int low = 0;
    int high = n - 1;
    int floor_val = -1; // Default if no floor exists

    while (low <= high) {
        int mid = low + (high - low) / 2;

        // If elements matches x, it's the exact floor
        if (arr[mid] == x) {
            return arr[mid];
        }
        // If element is smaller than x, save it and search right
        else if (arr[mid] < x) {
            floor_val = arr[mid];
            low = mid + 1;
        }
        // If element is larger than x, search left
        else {
            high = mid - 1;
        }
    }
    return floor_val;
}

int main() {
    int n, x;

    // Read size of array
    if (scanf("%d", &n) != 1) return 0;

    // Allocate memory dynamically for n integers
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1; // Memory allocation failed
    }

    // Read n elements line-by-line
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0;
        }
    }

    // Read the target value x
    if (scanf("%d", &x) != 1) {
```

```
51 ▾    if (scanf("%d", &x) != 1) {
52          free(arr);
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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